RESEARCH PROFILE

Yujie Huang

Second-year master's student · LLM / Agents / AI for Science

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Education

Fujian University of Technology — Master's student · Year 2

Research areas: large language models (LLMs) and AI agents.

Selected Publications

01 / Thinking in Flow

ICML 2026 · Spotlight

Thinking in Flow: A Dissipative Stabilization Operator for Robust Autoregressive Reasoning

Yujie Huang, Wenwu He, Zhuo-Xu Cui

- First author. Models continuous thought states with dissipative dynamics, using a lightweight Neural ODE controller and selective gates to stabilize long-horizon autoregressive reasoning.
- Analyzes well-posedness, dissipativity, and incremental stability; studies reasoning robustness using the Llama and Qwen model families.

https://icml.cc/virtual/2026/poster/65866

02 / HER-PDE

arXiv preprint

Hypothesize, Evaluate, Refine: A Scientific Agent for Open-Form PDE Discovery with Unknown Spatial Coefficient Fields

Yujie Huang, Wenwu He, Zhuoer Lin, Congcong Liu, Dong Liang, Zhuo-Xu Cui

- First author. Couples expression-tree search by a scientific agent with HEI numerical evaluation to jointly discover open-form PDE structures and unknown spatial coefficient fields.
- Updated manuscript: 70/105 exact structure recoveries and 94.3% Pass@3 across seven two-dimensional PDE families. Technical-repeat validation on real microbial trajectories reduces continuous-prediction error by 15.7% versus a refitted linear reference.

https://arxiv.org/abs/2608.27475

Honors & Recognition

ICML 2026 Spotlight · Thinking in Flow — 2026

Best Paper Nomination · UMI 2026 Youth Forum — 2026

First-Class Scholarship · Top 5% — 2026

Research Focus

LLM reasoning & robustness / Scientific agents / Data-driven equation discovery